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Techniques for providing speed limit information

Abstract

Embodiments of the present disclosure set forth techniques for providing speed limit information to a vehicle. In an aspect, a computer-implemented method includes receiving, from a vehicle, navigation information associated with the vehicle; determining one or more predicted positions associated with the vehicle based on the navigation information; determining one or more candidate map regions associated with the vehicle based on the one or more predicted positions; associating corresponding speed limit information with each of the one or more candidate map regions; and transmitting, to the vehicle, the one or more candidate map regions; wherein the one or more candidate map regions are usable by the vehicle to provide speed limit information to a driver of the vehicle.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the priority benefit of U.S. Provisional Application No. 63/224,180, filed on Jul. 21, 2021, titled “TECHNIQUES FOR PROVIDING SPEED LIMIT INFORMATION.” The subject matter of this application is incorporated by reference herein in its entirety.

BACKGROUND

Field of the Various Embodiments

(1) The various embodiments relate generally to vehicle information systems, and more specifically, to techniques for providing speed limit information.

DESCRIPTION OF THE RELATED ART

(2) Intelligent speed assistance (ISA) is a driver assistance technology that can be implemented by vehicles to provide speed limit information (SLI) to an operator (e.g., driver) of a vehicle, and in some cases, to actively intervene to adjust a speed of the vehicle. Certain jurisdictions, most notably the European Union, have adopted regulations that mandate ISA in new vehicles.

(3) A typical ISA implementation includes a camera on the vehicle for recognizing speed limit signs on roads and on-board correlation of a current location of the vehicle to a map database that includes speed limit information. Additionally, the map database needs periodic updating to maintain reliability of the ISA implementation. A drawback of this approach is that this approach is resource intensive. This approach entails inclusion of a navigational system and a map database in the vehicle. However, many existing or new entry level vehicles are unable to support a navigational system without significant additional expense. Further, this approach requires a high level of computational power for the location-to-map correlation process, which can significantly increase the cost of an entry level vehicle. Accordingly, this approach can make entry level vehicles more expensive.

(4) As the foregoing illustrates, what is needed are more effective techniques for providing speed limit information in vehicles.

SUMMARY

(5) One embodiment sets forth a computer-implemented method comprising receiving, from a vehicle, navigation information associated with the vehicle; determining one or more predicted positions associated with the vehicle based on the navigation information; determining one or more candidate map regions associated with the vehicle based on the one or more predicted positions; associating corresponding speed limit information with each of the one or more candidate map regions and transmitting, to the vehicle, the one or more candidate map regions; wherein the one or more candidate map regions are usable by the vehicle to provide speed limit information to a driver of the vehicle.

(6) One embodiment sets forth a computer-implemented method comprising transmitting navigation information associated with a vehicle to a server; receiving, at the vehicle and from the server, map region information associated with a plurality of candidate map regions; selecting a first candidate map region included in the plurality of candidate map regions based on a location of the vehicle; and causing one or more operations to be performed at the vehicle based on speed limit information associated with the first candidate map region.

(7) Further embodiments provide, among other things, one or more non-transitory computer-readable media and systems configured to implement the methods set forth above.

(8) At least one technical advantage of the disclosed techniques relative to the prior art is that, with the disclosed techniques, a vehicle can obtain and output speed limit information using less computational resources, including less processing power and data storage in the vehicle, compared to prior approaches. Accordingly, intelligent speed assistance can be implemented in a vehicle at lower cost, including for example using a simpler, less expensive, lower power processor. Additionally, with the disclosed techniques, a vehicle does not need to have access to a navigational database and does not need to update the navigational database, which reduces the

memory and/or storage required in the vehicle, the bandwidth needed to receive the updates, and/or service time at a vehicle dealer to install updates. These technical advantages provide one or more technological improvements over prior art approaches.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) So that the manner in which the above recited features of the various embodiments can be understood in detail, a more particular description of the inventive concepts, briefly summarized above, may be had by reference to various embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of the inventive concepts and are therefore not to be considered limiting of scope in any way, and that there are other equally effective embodiments.
- (2) FIG. 1 is a block diagram of a vehicle-based computer system configured to implement one or more aspects of the various embodiments;
- (3) FIG. 2 illustrates aspects of the computer system of FIG. 1 in further detail, according to one or more aspects of the various embodiments;
- (4) FIG. 3 illustrates an example road layout and associated map regions, according to one or more aspects of the various embodiments;
- (5) FIG. 4 is a flow chart of method steps for providing speed limit information to a vehicle, according to one or more aspects of the various embodiments; and
- (6) FIG. 5 is a flow chart of method steps for determining speed limit information for a vehicle location, according to one or more aspects of the various embodiments.

DETAILED DESCRIPTION

- (7) In the following description, numerous specific details are set forth to provide a more thorough understanding of the various embodiments. However, it will be apparent to one of skilled in the art that the inventive concepts may be practiced without one or more of these specific details.
- (8) FIG. 1 illustrates a block diagram of a vehicle-based computer system **100** configured to implement one or more aspects of the various embodiments. As shown, vehicle-based computer system **100** includes, without limitation, computing device **180**, sensor(s) **120**, input/output (I/O) device(s) **130**, location module **138**, network **160**, and server(s) **170**. Computing device **180** is located in a vehicle and includes, without limitation, a processing unit **102**, I/O device interface **104**, network interface **106**, interconnect (bus) **112**, storage **114**, and memory **116**. Memory **116** stores an Intelligent Speed Assist (ISA) client application **150**. Network **160** couples computing device **180** to server(s) **170**. Each of server(s) **170** include map data **172** and an ISA server application **174**. Processing unit **102**, I/O device interface **104**, network interface **106**, storage **114**, and memory **116** can be communicatively coupled to each other via interconnect **112**.
- (9) Computing device **180** includes processing unit **102** and memory **116**. Computing device **180** can be a device that includes one or more processing units **102**, such as a system-on-a-chip (SoC). In some embodiments, computing device **180** is implemented within a vehicle. For example, computing device **180** could be included in a head unit and/or other in-vehicle computing system.
- (10) Processing unit **102** includes a central processing unit (CPU), a digital signal processing unit (DSP), a microprocessor, an application-specific integrated circuit (ASIC), a neural processing unit (NPU), a graphics processing unit (GPU), a field-programmable gate array (FPGA), and/or the like, or a combination of multiple processing units, such as a CPU configured to operate in conjunction with a GPU. Processing unit **102** generally comprises a programmable processor that executes program instructions to manipulate input data. In some embodiments, processing unit **102** includes any number of processing cores, memories, and other modules for facilitating program execution. For example, processing unit **102** could receive input from a user via I/O devices **130** and generate

pixels for display on an I/O device **130** (e.g., a display device **134**). In some embodiments, processing unit **102** can be configured to execute ISA client application **150** in order to provide speed limit information to the driver of the vehicle and/or perform one or more operations based on the speed limit information.

(11) In various embodiments, storage **114** includes non-volatile storage for applications, software modules, and data, and can include fixed or removable disk drives, flash memory devices, and CD-ROM, DVD-ROM, Blu-Ray, HD-DVD, or other magnetic, optical, solid state storage devices, and/or the like. For example, ISA client application **150** could be stored in storage **114**, and then loaded into memory **116** as needed.

(12) In various embodiments, memory **116** includes a memory module or collection of memory modules. Memory **116** generally comprises storage chips such as random access memory (RAM) chips that store application programs and data for processing by processing unit **102**. Processing unit **102**, I/O device interface **104**, and network interface **106** are configured to read data from and write data to memory **116**. ISA client application **150** can be loaded from storage **114** into memory **116**. While in memory **116**, ISA client application **150** is executed by processing unit **102** to implement the functionality described in the present disclosure.

(13) In various embodiments, processing unit **102** is configured to store values in storage **114** and/or memory **116**, and/or retrieve values stored in storage **114** and/or memory **116**. For example, processing unit **102**, executing ISA client application **150**, stores speed limit information and/or map region information in storage **114** between the receipt of updates from ISA server application **174**.

(14) Sensor(s) **120** include one or more devices that perform measurements and/or acquire data related to vehicle operation, such as vehicle speed information and/or the like. In some embodiments, sensor(s) **120** are coupled to computing device **180** via a CAN bus and/or other in-vehicle communication network. Examples of sensors **120** include, without limitation, accelerometers, gyroscopes, magnetometers, light sensors, thermometers, inertial measurement units, compass, vehicle tilt sensors, crankshaft sensors, rain sensors, engine temperature sensors, intake-air temperature sensors, coolant temperature sensors, outdoor temperature sensors, accelerator pedal sensors, wheel speed sensors, air conditioning on/off sensors, air-mass sensors, steering angle sensors, road speed sensors, parking sensors, lambda sensors, seat belt sensors, transmission range sensors, knock sensors, and/or the like.

(15) Location module **138** includes hardware and/or software components for determining a geographic location of computing device **180** (e.g., a current location of the vehicle). In some examples, location module **138** determines a location of computing device **180** via acquisition of geolocation data (e.g., from a global navigation satellite system, such as a global positioning system (GPS), GNSS, Glonass, Galileo, Beidou, etc.) and/or determination of location based on sensor data from sensors **120** (e.g., dead reckoning).

(16) I/O device(s) **130** includes devices capable of receiving input, such as a keyboard, a mouse, a touch-sensitive screen, a microphone and other input devices for providing input data to computing device **180**. In various embodiments, I/O device(s) **130** includes devices capable of providing output, such as a display screen, one or more speakers (e.g., a loudspeaker array for spatial sound generation, ultrasound speakers for highly directional sound generation), haptic devices, touchless haptic devices (e.g., for creating haptic sensations at a distance), and/or the like. One or more of I/O devices **130** can be incorporated in computing device **180** or may be external to computing device **180**. I/O devices **130** interface with computing device **180** via I/O devices interface **104**. In some embodiments, computing device **180** and/or one or more I/O device(s) **130** may be components of an infotainment system implemented in a vehicle.

(17) In various embodiments, I/O devices **130** include one or more audio speakers **132**, one or more display devices **134**, and/or one or more input devices **136**. Examples of display devices **134** include, without limitation, LCD displays, LED displays, heads-up displays, head-mounted

displays, and touch-sensitive screens. Examples of input devices **136** include, without limitation, buttons, knobs, dials, a touchpad, a touch-sensitive screen, joysticks, and a microphone.

Additionally, I/O devices **130** can include other devices capable of receiving input and/or output.

(18) Network **160** enables communications between computing device **180** and other devices in network **160** via wired and/or wireless communications protocols, satellite networks, V2X networks, including Bluetooth, Bluetooth low energy (BLE), wireless local area network (WiFi), cellular protocols, and/or near-field communications (NFC). Network **160** is any technically feasible type of communications network that allows data to be exchanged between computing device **180** and other devices in network **160** (e.g., server(s) **170**). For example, network **160** could include a wide area network (WAN), a local area network (LAN), a wireless network (e.g., a Wi-Fi network, a cellular data network), and/or the Internet, among others. Computing device **180** connects with network(s) **160** via network interface **106**. In some embodiments, network interface **106** is hardware, software, or a combination of hardware and software, that is configured to connect to and interface with network(s) **160**.

(19) Server(s) **170** include any type of cloud computing system, cloud-based storage, distributed computing system, or other networked computing device or system. Although not shown, each of server(s) **170** may include processing unit(s), memory, storage, a network interface, and/or an I/O device interface similar to those described with respect to computing device **180**. Server(s) **170** include map data **172** describing road segments and corresponding speed limit information. In some embodiments, map data **172** includes a database of road segments and corresponding information (e.g., road name, geolocation of the road segment, road signs along the road segment), including corresponding speed limit information, and/or the like. Map data **172** is periodically updated to capture changes in road configurations, speed limits, and/or the like. In various embodiments, map data **172** remains at server(s) **170**, and computing device **180** does not need to access map data **172** (e.g., computing device **180** is not required to store map data **172**).

Accordingly, computing device **180** does not need to receive map data updates or be serviced to have map data updates installed. Server(s) **170** further includes an ISA server application **174** that receives navigation information from ISA client application **150**, determines speed limit information from the navigation information and map data **172**, and provides the speed limit information to ISA client application **150** as described in further detail below. In some embodiments, ISA server application **174** executes in one or more virtual machines.

(20) FIG. 2 illustrates aspects of vehicle-based computer system **100** in further detail, according to one or more aspects of the various embodiments. Vehicle-based computer system **100**, as described above, includes ISA client application **150** and ISA server application **174**. As shown, ISA client application **150** includes, without limitation, navigation information module **202**, map region selection module **214**, and speed limit information operation module **216**. As shown, ISA server application **174** includes, without limitation, position prediction module **206** and map region module **210**.

(21) In operation, navigation information module **202** determines navigation information **204** associated with a vehicle that includes computing device **180**. In some embodiments, navigation information **204** includes a geolocation of the vehicle, a travelling direction of the vehicle, and a speed of the vehicle. Alternatively, the traveling direction and the speed of the vehicle can be combined as a velocity of the vehicle. The geolocation of the vehicle is a location of the vehicle, which can be expressed as coordinates (e.g., latitude and longitude). Navigation information module **202** obtains the geolocation via location module **138**. The travelling direction is a direction toward which the vehicle is travelling or facing, which can be expressed as a cardinal direction or an angular degree relative to north. Navigation information module **202** obtains the direction via location module **138** and/or sensors **120** (e.g., a compass). A speed of the vehicle is the speed at which the vehicle is travelling. Navigation information module **202** can obtain the vehicle speed via a speedometer on the vehicle. ISA client application **150** transmits navigation information **204**

to ISA server application **174** on server(s) **170** via a network, such as network **160**. In some embodiments, navigation information module **202** determines and transmits navigation information **204** periodically based on time (e.g., every minute) or distance travelled (e.g., every 100 meters travelled), at a next update time determined by ISA server application **174**, and/or the like.

(22) Position prediction module **206** receives navigation information **204** from computer device **180** and processes navigation information **204** to determine a predicted position **208** of the vehicle. Predicted position **208** represents a geolocation position (e.g., coordinates) at which the vehicle is predicted to be located in the near future based on the geolocation, the travelling direction, and the speed of the vehicle. In some embodiments, position prediction module **206** first matches the geolocation of the vehicle to a map area (e.g., a road segment); the matched map area represents where on the map the vehicle is currently travelling. Position prediction module **206** then determines, based on the travelling direction and speed on the matched map area, including the road segments included in the matched map area, one or more predicted positions **208** based on whether the vehicle continues traveling on the current road segment or takes one or more turns onto intersecting road segments. In some embodiments, position prediction module **206** determines a plurality of predicted positions **208**. Each of the plurality of predicted positions **208** corresponds a candidate prediction of a location on the map to which the vehicle will travel based on the geolocation, travelling direction, vehicle speed, and the matched map area.

(23) Position prediction module **206** can determine predicted position(s) **208** using any suitable and technically feasible approach. For example, position prediction module **206** can implement a machine learning-based technique that determines predicted position(s) **208** using a trained model of driving behaviors. In some embodiments, position prediction module **206** determines a predicted position(s) **208** along with an associated confidence level. For example, position prediction module **206** could determine a plurality of predicted positions **208** and a confidence level for each determined predicted position.

(24) In some embodiments, position prediction module **206** determines predicted position(s) **208** for a certain amount of time or distance of travel from of the vehicle geolocation, given the travelling direction and speed. For example, position prediction module **206** could determine predicted position(s) **208** that is 5 seconds ahead of travel from the vehicle geolocation. As another example, position prediction module **206** could determine a predicted position(s) **208** that is 100 meters ahead of travel from the vehicle geolocation. In some embodiments, the predict-ahead time or distance for predicted position(s) **208** is between 1-5 seconds or 1-100 meters, respectively. In some embodiments, the predict-ahead time or distance is configurable (e.g., by the driver of the vehicle). Further, in some embodiments, position prediction module **206** dynamically selects a predict-ahead time or distance based on navigation information **204** and information derived from navigation information **204**. For example, position prediction module **206** could determine, based on the geolocation, that the type of road currently travelled is a long-stretch road (e.g., a highway) with few or no intersecting roads, and thus could then select a longer predict-ahead time or distance based on a lower possibility that the vehicle will deviate from the road currently being travelled on. As another example, position prediction module **206** could determine, based on the geolocation, that the type of road currently travelled is a city street with intersecting roads in proximity, and thus could then select a shorter predict-ahead time or distance based on a higher possibility that the vehicle will deviate from the road currently travelled.

(25) In some embodiments, position prediction module **206** also determines a next update time based on navigation information **204** and map data **172**. The next update time is a next earliest time that ISA client application **150** is suggested to wait to transmit updated navigation information **204** to ISA server application **174** in order to have ISA server application **174** determine new predicted position(s) **208**. The next update time can be expressed as a timestamp of when ISA client application **150** could next send navigation information **204**, or an amount of time for ISA client application **150** to wait after the last transmission of navigation information **204** to send updated

navigation information **204**.

(26) Map region module **210** receives predicted position(s) **208** from position prediction module **206**. Map region module **210** determines one or more map regions **220** based on predicted position(s) **208** and map data **172**. In some embodiments, map region module **210** determines a map region **220** for each of a plurality of predicted positions **208**. A map region **220** represents a map area (e.g., one or more road segments) that has a common, specified speed limit for that area. In some embodiments, a map region **220** is a map area that is specified as a simple polygonal (convex or concave) area defined by at least three vertices, where the vertices are geolocations (e.g., location coordinates). Map region(s) **220** correspond to map areas that are candidates for being a part of the vehicle's actual path based on predicted position(s) **208**.

(27) In some embodiments, map region module **210** identifies, in map data **172**, one or more road segments that are in proximity to predicted position(s) **208**. Map region module **210** also retrieves speed limit information (e.g., speed limit values) of those road segments from map data **172**. Map region module **210** generates one or more map regions **220** covering those road segments. A map region **220** covers one or more road segments with a common speed limit, and can cover travel in one or both directions of a road segment. Map region module **210** generates a map region **220** for a road segment on which a predicted position **208** is located. In some embodiments, map region module **210** generates a map region **220** for each of one or more road segments. In some embodiments, map region module **210** can generate a map region **220** that covers multiple road segments (e.g., multiple intersecting or connected road segments) that have a common speed limit. Map regions for each of the one or more predicted positions are described in further detail below in FIG. 3.

(28) Map region module **210** also determines one or more map regions, amongst map regions **220**, for special cases. In some embodiments, map region module **210** determines that a predicted position **208** is within or in proximity of a tunnel (e.g., the tunnel is located from a predicted position **208** at a distance that may be reachable before the next time ISA client application **150** transmits navigation information **204**). Based on the predicted position **208** being determined to be within or in proximity of a tunnel, map region module **210** determines a map region, amongst map regions **220**, that covers the tunnel. A map region **220** corresponding to a tunnel (hereinafter a "tunnel map region") includes an indication that the tunnel map region corresponds to a tunnel, but is otherwise similar to other map regions **220** described above (e.g., includes a polygonal area defined by geolocation vertices, etc.). A tunnel map region provides speed limit information for a road segment in a tunnel, where the vehicle is unable to obtain a geolocation via location module **138** and is unable to communicate with server(s) **170**.

(29) In some embodiments, map region module **210** determines a default or fallback map region amongst map regions **220**. The default map region covers a map area surrounding predicted position **208** and corresponds to a large area that encompasses a type of road (e.g., rural, urban, highway) with a common speed limit. The default map region has a speed limit that is associated with the country, region, and/or type of road where the vehicle is travelling, and thus can cover a larger map area than other map regions **220** determined by map region module **210**. A default map region provides default speed limit information when the geolocation of the vehicle does not fall within the areas of other map regions **220**. More generally, a default map region is similar to other map regions **220** (e.g., being a polygonal region with location coordinates as vertices, has a speed limit for the region), with possible differences as described above.

(30) ISA server application **174** transmits map region information **212** to ISA client application **150** on computing device **180** via a network, such as network **160**. Map region information **212** includes data indicating map region(s) **220** generated by map region module **210**. For a given map region **220**, map region information **212** includes location coordinates corresponding to polygonal vertices of the given map region **220**, and a speed limit value (e.g., 50 miles per hour, 40 km per hour, etc.) for the given map region **220**. In some embodiments, ISA server application **174** also

transmits a next update time, determined by position prediction module **206**, to ISA client application **150**. In some embodiments, map region information **212** also includes data indicating a default map region and/or a tunnel map region amongst map regions **220**. Data indicating a tunnel map region includes an indication that the tunnel map region corresponds to a tunnel. ISA client application **150** receives map region information **212** from ISA server application **174**. In some embodiments, ISA client application **150** caches map region information **212** (e.g., in memory **116**). In some embodiments, map regions **220** do not overlap with each other, other than a default map region possibly overlapping with one or more other map regions **220**.

(31) Map region selection module **214** selects a map region amongst map regions **220** included in map region information **212** based on the geolocation of the vehicle. That is, map region selection module **214** determines a geolocation of the vehicle via location module **138** and compares that vehicle geolocation to the map regions **220** included in map region information **212** to determine whether the vehicle geolocation is within one of the map regions. Map region selection module **214** performs the comparison of the vehicle geolocation to a given map region **220** by comparing the vehicle geolocation to the polygonal region of the map region. If the vehicle geolocation is not within the polygonal region of a map region, then that map region is not selected. If the vehicle geolocation is within the polygonal region of a map region, then map region selection module **214** selects that map region. If map region selection module **214** is unable to determine a geolocation of the vehicle, then map region selection module **214** selects a tunnel map region if one is included in map region information **212** (under the assumption that the location module **138** is unable to determine the geolocation because the vehicle is in a tunnel). If map region selection module **214** is unable to determine a geolocation of the vehicle and no tunnel map region is included in map region information **212**, or if the vehicle geolocation is not in the polygonal region of any of map regions **220** included in map regions **220** other than the default map region, then map region selection module **214** selects the default map region.

(32) After selecting a map region amongst map regions **220**, map region selection module **214** passes the speed limit value for the selected map region to SLI operation module **216**. SLI operation module **216** performs one or more operations and/or causes one or more operations to be performed based on the speed limit value. In some examples, SLI operation module **216** outputs the value to an in-vehicle user interface (e.g., graphical user interface, heads-up display, text-to-speech system) for output to the user (e.g., displayed on display device **134**, read out in speech on speakers **132**). In some examples, SLI operation module **216** additionally intervenes in operation of the vehicle based on the speed limit value. For example, if the current speed of the vehicle is determined to exceed the speed limit value (e.g., exceed the speed limit value by any amount or by at least a minimum amount), SLI operation module **216** can instruct a braking system in the vehicle to apply brakes in order to decrease the vehicle speed.

(33) After ISA client application **150** receives map region information **212**, ISA client application **150** waits for the next update time to determine and transmit new navigation information **204** to ISA server application **174**. In some examples, the next update time is determined by ISA server application **174** and transmitted along with map region information **212**, is a configured update period, and/or the like.

(34) FIG. 3 illustrates an example road layout and associated map regions, according to one or more aspects of the various embodiments. As described above, ISA server application **174** determined one or more map regions **220** for a vehicle based on one or more predicted positions **208** of the vehicle. FIG. 3 illustrates a road layout **300** and examples map regions that could be determined for road layout **300**. As shown, FIG. 3 includes a vehicle **302** at its current location, travelling on road segment **304** in a first direction. A road segment **306** branches from road segment **304** and vehicle **302** can turn right on road segment **306** or continue straight on road segment **304**. FIG. 3 further shows an upcoming intersection on road segment **304** where vehicle **302** can turn right into road segment **310** or turn left into road segment **312**.

(35) ISA client application **150** in vehicle **302** determines navigation information **204** for vehicle **302**, including a geolocation, a travelling direction (the first direction), and a speed of the vehicle. ISA client application **150** transmits navigation information **204** for vehicle **302** to ISA server application **174**. Based on navigation information **204** for vehicle **302** and map data **172** corresponding to road layout **300**, position prediction module **206** determines that the predicted position of vehicle **302** could be on any of the road segments **304**, **306**, **310**, and/or **312**. Based on these predicted positions and map data **172** corresponding to road layout **300**, map region module **210** can generate map regions **308** for vehicle **302**. Map regions **308** include a map region **308-1** for a portion of road segment **304** in front of vehicle **302**, map region **308-2** for road segment **306**, and map regions **308-3** and **308-4** for road segments **310** and **312**, respectively. As shown, map regions **308** are shaped as polygons (e.g., rectangle, parallelogram), and each map region **308** is associated with a speed limit value. Additionally, map region module **210** can generate a default map region (not shown) in FIG. **3** based on road layout **300**, and if a tunnel is in map regions **308** that map region is designated as a tunnel map region. ISA server application **174** transmits map region information **212** indicating map regions **308** to ISA client application **150** in vehicle **302**. ISA client application **150** in vehicle **302** selects one of map regions **308** based on a current geolocation of vehicle **302**.

(36) After ISA client application **150** receives map region information **212** indicating map regions **308**, ISA client application **150** determines a current geolocation of vehicle **302** and selects one of map regions **308** based on the current geolocation. As shown, vehicle **302** has traveled further on road segment **304** to geolocation **314**. ISA client application **150** determines, via location module **138**, that vehicle **302** is located at geolocation **314**. ISA client application **150** then compares geolocation **314** to map region **308** to determine the map region in which geolocation **314** is located, if any. ISA client application **150** determines that geolocation **314** is located in map region **308-1**, and in response selects map region **308-1**. ISA client application **150** performs one more operations based on the speed limit information associated with map region **308-1** (e.g., output the speed limit value for map region **308-1**, check if vehicle **302** is exceeding the speed limit value for map region **308-1**). Because map regions **308** are polygonal regions with defined boundaries, comparing a vehicle geolocation to map regions **308** and determining whether the vehicle geolocation is in any of map regions **308** is an operation that uses very little computing resources (e.g., processing power) compared to other approaches, such as comparing a vehicle geolocation to a map database to find speed limit information for the vehicle geolocation.

(37) FIG. **4** is a flow chart of method steps for providing speed limit information to a vehicle, according to one or more aspects of the various embodiments. Although the method steps are described with respect to the systems of FIGS. **1** and **2**, persons skilled in the art will understand that any system configured to perform the method steps, in any order, falls within the scope of the various embodiments. Additionally, although the method steps are described using the road layout **300** of FIG. **3** as an example, persons skilled in the art will understand that the method steps of method **400** can be used with any road layout.

(38) As shown, a method **400** begins at step **402**, where an ISA server application, such as ISA server application **174**, obtains navigation information associated with a vehicle. The ISA server application receives navigation information, such as navigation information **204**, from an ISA client application in a vehicle, such as ISA client application **150**. The received navigation information includes a geolocation of the vehicle, a travelling direction of the vehicle, and a speed of the vehicle.

(39) At step **404**, the ISA server application determines one or more predicted positions of the vehicle based on the navigation information. The ISA server application determines predicted positions by, for example, determining possible travel paths from the vehicle geolocation at the vehicle speed and travelling direction. In some embodiments, the ISA server application determines a road layout in proximity of the vehicle geolocation and determines one or more predicted

positions of the vehicle in road layout given the speed and travelling direction.

(40) At step **406**, the ISA server application determines one or more candidate map regions based on the one or more predicted positions of the vehicle. As an example, when using the road layout **300**, the ISA server application determines the one or more map regions **308** for the one or more predicted positions of the vehicle. In some examples, the ISA server application also determines a default map region that can be used if the vehicle geolocation is not located in any of the other determined map regions. Additionally, if the ISA server application determines that one of the predicted positions matches a road segment in a tunnel, then the ISA server application generates a tunnel map region covering the tunnel road segment.

(41) At step **408**, the ISA server application transmits information associated with the one or more candidate map regions to the vehicle. The ISA server application generates map region information that includes information indicating the map regions and transmits the map region information to the ISA client application on the vehicle via a network, such as network **160**. The map region information **212** data specifying the boundaries (e.g., defined by vertex coordinates) and speed limit values of the map regions determined in step **406**. As an example, when using the road layout **300**, the ISA server application would generate map region information that includes data specifying the boundaries and speed limit values of each of map regions **308**, and would transmit the map region information to vehicle **302**. Method **400** then returns to step **402**, where the ISA server application obtains new navigation information associated with the vehicle.

(42) FIG. 5 is a flow chart of method steps for determining speed limit information for a vehicle location, according to one or more aspects of the various embodiments. Although the method steps are described with respect to the systems of FIGS. 1-3, persons skilled in the art will understand that any system configured to perform the method steps, in any order, falls within the scope of the various embodiments. Additionally, although the method steps are described using the road layout **300** of FIG. 3 as an example, persons skilled in the art will understand that the method steps of method **500** can be used with any road layout.

(43) As shown, a method **500** begins at step **502**, where an ISA client application, such as ISA client application **150**, transmits navigation information associated with a vehicle, such as navigation information **204**, to a server. The ISA client application in the vehicle determines navigation information for the vehicle and transmits the navigation information to an ISA server application at a server, such as ISA server application **174** at server(s) **170**. The navigation information includes a geolocation of the vehicle, a travelling direction of the vehicle, and a speed of the vehicle.

(44) At step **504**, the ISA client application receives, from the ISA server application at the server, information associated with a plurality of candidate map regions, including speed limit information associated with the candidate map regions. Each of the map regions includes data specifying the respective boundaries (e.g., defined by vertex coordinates) and speed limit values of the map regions. The candidate map regions are determined by the ISA server application based on the navigation information transmitted in step **502**. As an example using map layout **300** when the vehicle with the ISA client application is vehicle **302**, the plurality of map regions are the map regions **308**.

(45) At step **506**, the ISA client application selects a first candidate map region included in the plurality of candidate map regions based on a vehicle location. The ISA client application determines a geolocation of the vehicle and compares the determined geolocation to the boundaries of the map regions received in step **504**. The ISA client application selects a map region that includes the geolocation of the vehicle. For example, when the map regions are the map regions **308** and the ISA client application determines that the vehicle is at geolocation **314**, the ISA client application selects map region **308-1**. In some embodiments, if the ISA client application is unable to determine a geolocation of the vehicle, then the ISA client application selects a tunnel map region. In some embodiments, if the ISA client application determines that the geolocation of the

vehicle is not in the polygonal region of any of the plurality of candidate map regions other than the default map region, then the ISA client application selects the default map region.

(46) At step **508**, the ISA client application performs one or more operations based on speed limit information associated with the first candidate map region. For example, the ISA client application can output a speed limit value associated with the map region selected in step **506** visually (e.g., display on display **134**) and/or audibly (e.g., read out as speech via speakers **132**). The ISA client application can additionally determine whether the vehicle is exceeding the speed limit value and, if so, output an alert (e.g., a speeding alert) visually and/or audibly, and/or intervene in operation of the vehicle (e.g., command the brake system to apply the brakes). In an example, the ISA client application of vehicle **302** can display the speed limit value for map region **308-1** on display **134** in vehicle **302**. ISA client application **150** at vehicle **302** can also determine a speed of vehicle **302** and whether that speed exceeds the speed limit value for map region **308-1**. If the ISA client application determines that the speed exceeds the speed limit value, then the ISA client application can output an alert and/or intervene in operation of vehicle **302** to slow vehicle **302** down. Method **500** then returns to step **502**, where the ISA client application transmits new navigation information associated with the vehicle to the server. In some embodiments, the ISA client application waits for a next update time before transmitting the new navigation information.

(47) In sum, an approach for providing speed limit information to a vehicle includes a cloud-based approach. A vehicle provides navigation information, such as geolocation, speed, and travelling direction, to a cloud-based application. The cloud-based application determines, based on the received navigation information, one or more predicted positions for the vehicle. The cloud-based application determines, based on the one or more predicted positions, one or more map regions with corresponding speed limit information. The cloud-based application transmits information identifying the map regions and the speed limit information for those map regions to the vehicle. The vehicle then determines which map region the vehicle is located in and outputs the corresponding speed limit information.

(48) At least one technical advantage of the disclosed techniques relative to the prior art is that, with the disclosed techniques, a vehicle can obtain and output speed limit information using less computational resources, including less processing power and data storage in the vehicle, compared to prior approaches. Accordingly, intelligent speed assistance can be implemented in a vehicle at lower cost, including for example using a simpler, less expensive, lower power processor. Additionally, with the disclosed techniques, a vehicle does not need to have access to a navigational database and does not need to update the navigational database, which reduces the memory and/or storage required in the vehicle, the bandwidth needed to receive the updates, and/or service time at a vehicle dealer to install updates. These technical advantages provide one or more technological improvements over prior art approaches.

(49) 1. In some embodiments, a computer-implemented method comprises receiving, from a vehicle, navigation information associated with the vehicle; determining one or more predicted positions associated with the vehicle based on the navigation information; determining one or more candidate map regions associated with the vehicle based on the one or more predicted positions; associating corresponding speed limit information with each of the one or more candidate map regions; and transmitting, to the vehicle, the one or more candidate map regions; wherein the one or more candidate map regions are usable by the vehicle to provide speed limit information to a driver of the vehicle.

(50) 2. The method of clause 1, wherein the navigation information comprises a geolocation of the vehicle, and at least one of: a direction and a speed of the vehicle, or a velocity of the vehicle.

(51) 3. The method of clauses 1 or 2, wherein a first candidate map region included in the one or more candidate map regions is a polygonal region.

(52) 4. The method of any of clauses 1-3, wherein the first candidate map region includes a set of location coordinates corresponding to vertices of the polygonal region.

- (53) 5. The method of any of clauses 1-4, wherein the corresponding speed limit information for the first candidate map region indicates a speed limit for road segments within the first candidate map region.
- (54) 6. The method of any of clauses 1-5, further comprising determining that a tunnel is located within a first candidate map region; and designating the first candidate map region is a tunnel map region.
- (55) 7. The method of any of clauses 1-6, wherein determining one or more candidate map regions associated with the vehicle comprises determining a default map region.
- (56) 8. In some embodiments, one or more non-transitory computer-readable storage media include instructions that, when executed by one or more processors, cause the one or more processors to perform the steps of transmitting navigation information associated with a vehicle to a server; receiving, at the vehicle and from the server, map region information associated with a plurality of candidate map regions; selecting a first candidate map region included in the plurality of candidate map regions based on a location of the vehicle; and causing one or more operations to be performed at the vehicle based on speed limit information associated with the first candidate map region.
- (57) 9. The one or more non-transitory computer-readable storage media of clause 8, wherein the map region information comprises, for the first candidate map region, the speed limit information and a set of location coordinates corresponding to vertices of a polygonal region of the first candidate map region.
- (58) 10. The one or more non-transitory computer-readable storage media of clauses 8 or 9, wherein selecting the first candidate map region based on the location of the vehicle comprises determining that the location of the vehicle is included within the first candidate map region; and in response to the determination that the location of the vehicle is included within the first candidate map region, selecting the first candidate map region.
- (59) 11. The one or more non-transitory computer-readable storage media of any of clauses 8-10, wherein the plurality of candidate map regions is determined based on the navigation information associated with the vehicle.
- (60) 12. The one or more non-transitory computer-readable storage media of any of clauses 8-11, wherein the first candidate map region is a default map region, and selecting the first candidate map region included in the plurality of candidate map regions based on the location of the vehicle comprises determining that the location of the vehicle is outside of the plurality of candidate map regions other than the first candidate map region; and in response to determining that the location of the vehicle is outside of the plurality of candidate map regions other than the first candidate map region, selecting the default map region.
- (61) 13. The one or more non-transitory computer-readable storage media of any of clauses 8-12, wherein the first candidate map region is a tunnel map region, and selecting the first candidate map region included in the plurality of candidate map regions based on the location of the vehicle comprises determining that the location of the vehicle cannot be determined; and in response to determining that the location of the vehicle cannot be determined, selecting the tunnel map region.
- (62) 14. The one or more non-transitory computer-readable storage media of any of clauses 8-13, wherein the navigation information comprises a geolocation of the vehicle, and at least one of: a direction and a speed of the vehicle, or a velocity of the vehicle.
- (63) 15. The one or more non-transitory computer-readable storage media of any of clauses 8-14, wherein causing the one or more operations to be performed at the vehicle comprises at least one of: outputting a speed limit value included in the speed limit information associated with the first candidate map region, or intervening in operation of the vehicle based on the speed limit information.
- (64) 16. In some embodiments, a system implemented at a vehicle comprises a memory storing an application; and one or more processors that, when executing the application, is configured to

transmit navigation information associated with the vehicle to a server; receive, from the server, map region information associated with a plurality of candidate map regions; select a first candidate map region included in the plurality of candidate map regions based on a location of the vehicle; and cause one or more operations to be performed at the vehicle based on speed limit information associated with the first candidate map region.

(65) 17. The system of clause 16, wherein the navigation information comprises a geolocation of the vehicle, and at least one of: a direction and a speed of the vehicle, or a velocity of the vehicle.

(66) 18. The system of clauses 16 or 17, wherein the map region information comprises, for the first candidate map region, the speed limit information and a set of location coordinates corresponding to vertices of a polygonal region of the first candidate map region.

(67) 19. The system of any of clauses 16-18, wherein the one or more processors, when executing the application, is configured to determine that the location of the vehicle is included within the first candidate map region; and in response to determining that the location of the vehicle is included within the first candidate map region, select the first candidate map region.

(68) 20. The system of any of clauses 16-19, wherein the one or more processors, when executing the application, is configured to at least one of: output a speed limit value included in the speed limit information associated with the first candidate map region, or intervene in operation of the vehicle based on the speed limit information.

(69) Any and all combinations of any of the claim elements recited in any of the claims and/or any elements described in this application, in any fashion, fall within the contemplated scope of the present disclosure and protection.

(70) The descriptions of the various embodiments have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments.

(71) Aspects of the present embodiments may be embodied as a system, method or computer program product. Accordingly, aspects of the present disclosure may take the form of an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, etc.) or an embodiment combining software and hardware aspects that may all generally be referred to herein as a “module,” a “system,” or a “computer.” In addition, any hardware and/or software technique, process, function, component, engine, module, or system described in the present disclosure may be implemented as a circuit or set of circuits. Furthermore, aspects of the present disclosure may take the form of a computer program product embodied in one or more computer readable medium(s) having computer readable program code embodied thereon.

(72) Any combination of one or more computer readable medium(s) may be utilized. The computer readable medium may be a computer readable signal medium or a computer readable storage medium. A computer readable storage medium may be, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing. More specific examples (a non-exhaustive list) of the computer readable storage medium would include the following: an electrical connection having one or more wires, a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), an optical fiber, a portable compact disc read-only memory (CD-ROM), an optical storage device, a magnetic storage device, or any suitable combination of the foregoing. In the context of this document, a computer readable storage medium may be any tangible medium that can contain, or store a program for use by or in connection with an instruction execution system, apparatus, or device.

(73) Aspects of the present disclosure are described above with reference to flowchart illustrations and/or block diagrams of methods, apparatus (systems) and computer program products according

to embodiments of the disclosure. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer program instructions. These computer program instructions may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine. The instructions, when executed via the processor of the computer or other programmable data processing apparatus, enable the implementation of the functions/acts specified in the flowchart and/or block diagram block or blocks. Such processors may be, without limitation, general purpose processors, special-purpose processors, application-specific processors, or field-programmable gate arrays.

(74) The flowchart and block diagrams in the figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods and computer program products according to various embodiments of the present disclosure. In this regard, each block in the flowchart or block diagrams may represent a module, segment, or portion of code, which comprises one or more executable instructions for implementing the specified logical function(s). It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustration, and combinations of blocks in the block diagrams and/or flowchart illustration, can be implemented by special purpose hardware-based systems that perform the specified functions or acts, or combinations of special purpose hardware and computer instructions.

(75) While the preceding is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A computer-implemented method, comprising: receiving, by a server from a vehicle, navigation information associated with the vehicle; determining, by the server, one or more predicted positions associated with the vehicle based on the navigation information, wherein the one or more predicted positions corresponds to a plurality of road segments; acquiring, by the server from a data store storing map data, speed limit information for one or more groups of road segments included in the plurality of road segments; determining, for each group of the one or more groups of road segments, that the group of road segments share a common speed limit; generating, by the server, one or more candidate map regions associated with the vehicle from the one or more groups of road segments; and transmitting, by the server to the vehicle, the one or more candidate map regions; wherein the one or more candidate map regions are usable by the vehicle to provide speed limit information to a driver of the vehicle.
2. The method of claim 1, wherein the navigation information comprises a geolocation of the vehicle, and at least one of: a direction and a speed of the vehicle, or a velocity of the vehicle.
3. The method of claim 1, wherein a first candidate map region included in the one or more candidate map regions is a polygonal region.
4. The method of claim 3, wherein the first candidate map region includes a set of location coordinates corresponding to vertices of the polygonal region.
5. The method of claim 1, further comprising: determining that a tunnel is located within a first candidate map region; and designating the first candidate map region as a tunnel map region.
6. The method of claim 1, wherein determining the one or more candidate map regions associated with the vehicle comprises determining a default map region.

7. One or more non-transitory computer-readable storage media including instructions that, when executed by one or more processors, cause the one or more processors to perform the steps of: transmitting navigation information associated with a vehicle to a server; receiving, at the vehicle and from the server, map region information associated with a plurality of candidate map regions that the server selected based on the navigation information, wherein the server generates each of the candidate map regions by: acquiring, from a data store storing map data, speed limit information for a group of road segments, determining that the group of road segments share a common speed limit, and generating a candidate map region from the group of road segments; selecting a first candidate map region included in the plurality of candidate map regions based on a location of the vehicle; and causing one or more operations to be performed at the vehicle based on speed limit information associated with the first candidate map region.
8. The one or more non-transitory computer-readable storage media of claim 7, wherein: the map region information comprises, for each candidate map region: the speed limit information; and a set of location coordinates corresponding to vertices of a polygonal region of the candidate map region the steps further comprise: acquiring, from a data store storing map data, speed limit information for a group of road segments; determining that the group of road segments share a common speed limit; and generating the first candidate map region from the group of road segments.
9. The one or more non-transitory computer-readable storage media of claim 7, wherein selecting the first candidate map region based on the location of the vehicle comprises: determining that the location of the vehicle is included within the first candidate map region; and in response to the determination that the location of the vehicle is included within the first candidate map region, selecting the first candidate map region.
10. The one or more non-transitory computer-readable storage media of claim 7, wherein the plurality of candidate map regions is determined based on the navigation information associated with the vehicle.
11. The one or more non-transitory computer-readable storage media of claim 7, wherein the first candidate map region is a default map region, and selecting the first candidate map region included in the plurality of candidate map regions based on the location of the vehicle comprises: determining that the location of the vehicle is outside of the plurality of candidate map regions other than the first candidate map region; and in response to determining that the location of the vehicle is outside of the plurality of candidate map regions other than the first candidate map region, selecting the default map region.
12. The one or more non-transitory computer-readable storage media of claim 7, wherein the first candidate map region is a tunnel map region, and selecting the first candidate map region included in the plurality of candidate map regions based on the location of the vehicle comprises: determining that the location of the vehicle cannot be determined; and in response to determining that the location of the vehicle cannot be determined, selecting the tunnel map region.
13. The one or more non-transitory computer-readable storage media of claim 7, wherein the navigation information comprises a geolocation of the vehicle, and at least one of: a direction and a speed of the vehicle, or a velocity of the vehicle.
14. The one or more non-transitory computer-readable storage media of claim 7, wherein causing the one or more operations to be performed at the vehicle comprises at least one of: outputting a speed limit value included in the speed limit information associated with the first candidate map region, or intervening in operation of the vehicle based on the speed limit information.
15. A system implemented at a vehicle comprising: a memory storing an application; and one or more processors that, when executing the application, is configured to: transmit navigation information associated with the vehicle to a server; receive, from the server, map region information associated with a plurality of candidate map regions that the server selected based on the navigation information wherein the server generates each of the candidate map regions by: acquiring, from a data store storing map data, speed limit information for a group of road segments,

determining that the group of road segments share a common speed limit, and generating a candidate map region from the group of road segments; select a first candidate map region included in the plurality of candidate map regions based on a location of the vehicle; and cause one or more operations to be performed at the vehicle based on speed limit information associated with the first candidate map region.

16. The system of claim 15, wherein the navigation information comprises a geolocation of the vehicle, and at least one of: a direction and a speed of the vehicle, or a velocity of the vehicle.

17. The system of claim 15, wherein: the map region information comprises, for each candidate map region: the speed limit information; and a set of location coordinates corresponding to vertices of a polygonal region of the first candidate map region the server generates the first candidate map region by: acquiring, from a data store storing map data, speed limit information for a group of road segments; determining that the group of road segments share a common speed limit; and generating the first candidate map region from the group of road segments.

18. The system of claim 15, wherein the one or more processors, when executing the application, is configured to: determine that the location of the vehicle is included within the first candidate map region; and in response to determining that the location of the vehicle is included within the first candidate map region, select the first candidate map region.

19. The system of claim 15, wherein the one or more processors, when executing the application, is configured to at least one of: output a speed limit value included in the speed limit information associated with the first candidate map region, or intervene in operation of the vehicle based on the speed limit information.
